Remainder theorem

1

a $x = -3$ **b** $x = 8$ **c** $x = 6$ **d** $x = -10$

e $x = \frac{7}{6}$ **f** $x = \frac{1}{3}$ **g** $x = -\frac{1}{2}$ **h** $x = -\frac{5}{4}$

2

a -93 **b** 2100 **c** $\frac{125}{27}$ **d** -5

e 67 **f** $\frac{2205}{8}$ **g** $-\frac{41}{4}$ **h** $\frac{428}{81}$

3

a $k = -4$ **b** $k = 3$ **c** $k = 4$ **d** $k = -4$

e $k = -1$ **f** $k = 2$ **g** $k = 3$ **h** $k = -1$

4 $n = 6$

5

a 0

b In part (a), we found that $x^{99} + 1$ is divisible by $x + 1$. That is, $x^{99} + 1 = (x + 1) \times Q(x)$ (Equation 1) for some quotient $Q(x)$.
Substituting $x = 4$ into Equation 1, we get $4^{99} + 1 = 5 \times Q(x)$. Therefore, $4^{99} + 1$ is divisible by 5.

Factor theorem

6 Factoring

7 Expanding or multiplying

8 When it is written as a product of prime polynomials.

9 The factor theorem states that if f is a polynomial function and $f(c) = 0$, then $(x - c)$ is a factor of $f(x)$.

10 Using the factor theorem, $P(-2) = (-2)^4 + 3(-2)^3 - 7(-2)^2 - 27(-2) - 18 = 0$.

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a Using the factor theorem, $P(-4) = 2(-4)^3 + 8(-4)^2 + (-4) + 4 = 0$.

b $\left(2x^2 + 1\right)$

12

a Using the factor theorem, $P(5) = 10 - 17(5) + 8(5)^2 - (5)^3 = 0$.

b $\left(x^2 - 3x + 2\right)$

13

a $p = -3$

b $k = -84$

14 $Q\left(x\right) = x^3 - 4x^2 - 20x + 3$ because dividing $Q\left(x\right)$ by $\left(x + 3\right)$ gives a remainder of 0 and $Q\left(x\right)$ is the only polynomial such that $Q\left(-3\right) = 0$.

15

a $\left(x - 1\right)\left(x - 2\right)\left(x + 5\right)$

b $\left(x - 5\right)\left(x + 2\right)\left(x - 3\right)$

c $\left(x - 3\right)\left(x - 5\right)\left(x - 1\right)$

d $\left(x + 2\right)\left(x + 1\right)\left(x + 5\right)$

e $\left(2x - 5\right)\left(x - 1\right)\left(x - 2\right)$

f $\left(5x - 1\right)\left(x - 5\right)\left(x - 3\right)$

g $\left(2x + 3\right)\left(x - 2\right)\left(x + 1\right)$

h $\left(4x + 3\right)\left(x - 2\right)\left(x + 1\right)$

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